LIQUID LEVEL MEASUREMENT · VSG SERIES

MAGNETIC LEVEL INDICATOR

By-pass type analog level indicator for tanks and vessels, with no contact to the medium

The magnetic by-pass level indicator allows the liquid level in tanks to be monitored from outside by means of a magnetic float and a moving colour-changing indicator.

PN10 10 BAR	**100 °C / 250 °C** MAX. OPERATING TEMPERATURE
5800 mm ONE-PIECE LENGTH	**AISI 304 / 316** FULLY STAINLESS STEEL



VSG STANDARD

FEATURES

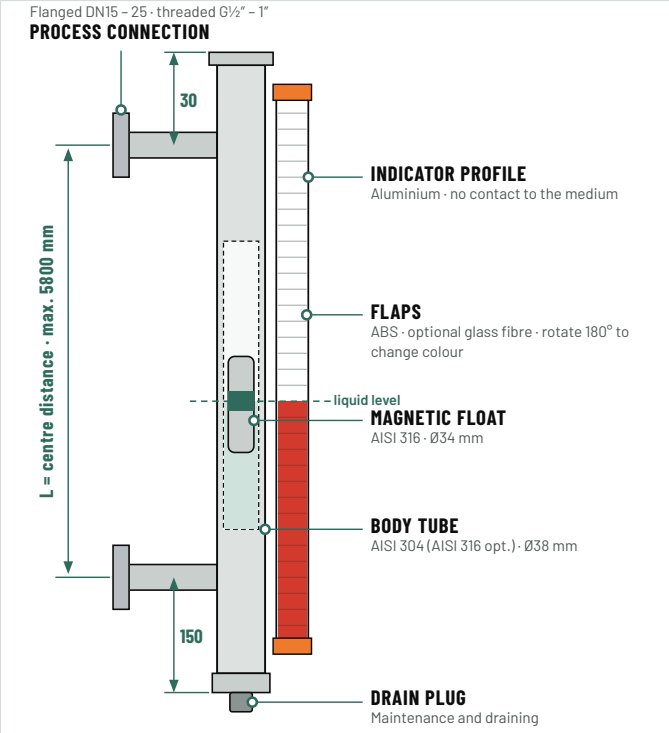
- Indicator profile with no contact to the medium
- Analog indicator easily readable from a distance
- Robust, shock-resistant body
- Dry contacts at any point with limit switches mounted on the indicator
- Easy installation with standard and swivel flanges
- Easy maintenance and installation with the valved model
- Custom manufacturing for different connections and lengths
- Level reading on the device or in percent, volume or metres with the scaled model

TECHNICAL SPECIFICATIONS

Pressure (Pmax)	PN10
Temperature (Tmax)	100 °C · optional max. 250 °C
Connection material	Carbon steel St 37.2 · stainless steel (optional)
Connection types	Threaded G½" · G¾" · G1" Flanged DN15 · DN20 · DN25
Max. length	5800 mm, one piece

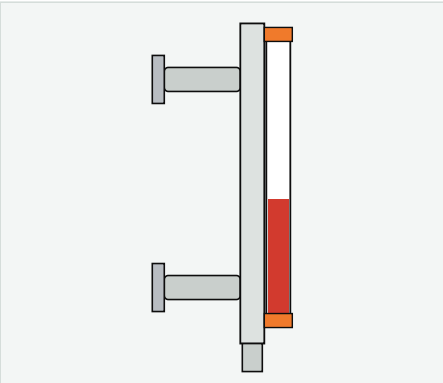
Float	AISI 316 · Ø34 mm
Tube	AISI 304 (AISI 316 optional) · Ø38 mm
Indicator profile	Aluminium
Flaps	ABS plastic · optional glass fibre (200 °C resistant)
Contact / signal	Dry contact via limit switch · 0–10 V / 4–20 mA (VSG 30)

DESIGN, OPERATING PRINCIPLE AND DIMENSIONS



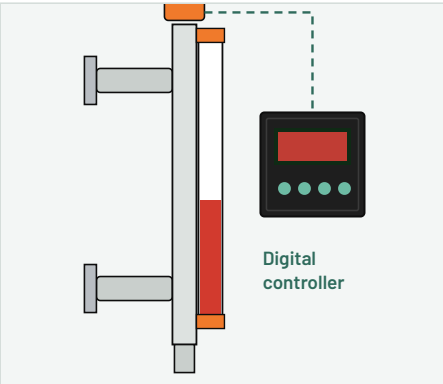
1. **Communicating vessels.** The indicator is connected to the tank as a by-pass via two process connections. The liquid level in the body tube is the same as in the tank.
2. **Magnetic float.** The float inside the tube floats on the liquid surface and moves with the level.
3. **Flap indicator.** The magnet in the float rotates the flaps in the profile by 180°. Below the level they show red, above white; the level can be read from a distance.
4. **Contact and signal.** Limit switches mounted on the profile provide dry contacts at the required points. In VSG 30 models the level is transmitted as an analog signal.

MODELS



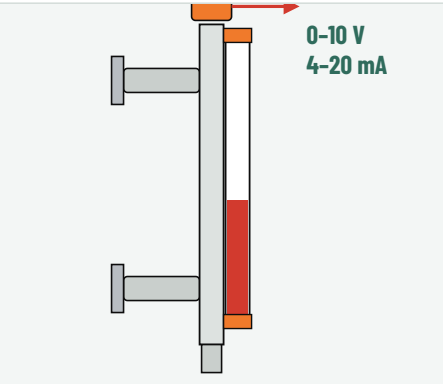
VSG STANDARD

Mounted on the tank or vessel as a by-pass. Level changes are monitored visually via the flaps. Limit switches can be fitted.



VSG 30

The level signal is transmitted to a digital controller, which displays the level in percent, volume, mass or metres.



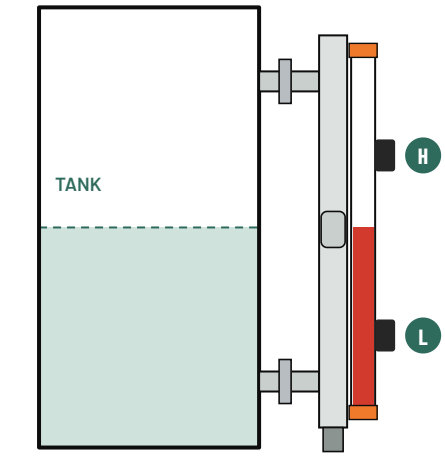
VSG 30 DIRECT OUTPUT

The level transmitter is integrated into the indicator. The level is sent directly to the automation system as a 0–10 V or 4–20 mA signal.

ACCESSORIES

ACCESSORY	DESCRIPTION	CONTACT / OUTPUT	VOLTAGE	TEMPERATURE
Limit switch	Mounted on the indicator profile, provides a dry contact at the required point. Connected via controller or directly to automation. Aluminium housing.	1.5 A · 50 W max. NO · NC (opt.)	250 VAC – 200 VDC	–60 ... +125 °C
High-temperature switch	Mounted on the indicator profile in high-temperature applications, provides a dry contact. Controller supported. Aluminium housing.	1.5 A · 50 W max. NO · NC (opt.)	250 VAC – 200 VDC	–60 ... +250 °C
Controller	Switches between the limit switches and the devices to be controlled. Up to 6 limit switches can be connected.	6 contact inputs 3 × 10 A NO + NC	220 VAC 50 Hz 24 VDC (opt.)	–
Digital controller	Used with VSG 11. The level is shown on the display and transmitted to automation via relay and analog outputs.	3 × 10 A NO + NC 4–20 mA · 0–10 V	220 VAC 50 Hz 24 VDC (opt.)	–
Analog indicator	Connected to VSG 11 or a digital controller; the liquid level is shown on the indicator display.	–	24 V	–

(opt.) = optional



H / L: limit switch points

INSTALLATION AND USE

01 The indicator is mounted vertically on two process connections on the tank side wall using standard or swivel flanges or threaded connections.

02 The centre distance (L) is specified in the order according to the tank connections; manufactured in one piece up to 5800 mm.

03 With the valved model, the process valves can be closed and the indicator serviced without draining the tank.

04 Limit switches are moved to the required level on the indicator profile, fixed, and provide dry contacts.

05 Sediment and residue in the tube are drained periodically through the drain plug.

CAUTION There must be no strong magnetic field or magnetic material near the indicator; it may affect correct flap rotation.

Custom manufacturing for other connections, lengths, pressures and temperatures. Orders and technical support: Karatay/Konya/TÜRKİYE · info@netflw.com